

Nexus Document Platform 4.2 — Database Design

Core Tables

documents stores document identity, tenant, title, content hash, classification, and state.

document_metadata stores extracted key/value metadata.

processing_jobs tracks asynchronous processing.

audit_events stores security and operational audit records.

Indexes

documents has indexes on tenant_id, status, created_at, and content_hash.

document_metadata has an index on document_id and metadata_key.

audit_events has an index on tenant_id, event_time, and event_type.

Transactions

Document creation and the initial processing job are committed atomically. Metadata updates use optimistic concurrency through the version column. A stale version causes the update to be rejected.

Connection Pooling

Backend services use a connection pool. The production maximum pool size is 50 per service instance. Connections should be released promptly using context-managed database operations.

Related Components

See schema.sql, connection_pooling.txt, database_config.json, and DatabaseService.cs.